

Module 10: Matrices and Gaussian Elimination – Prequiz

Name:

Date:

Four short questions. Translate freely between systems, augmented matrices, and row operations.

1. Write the augmented matrix for

$$\begin{cases} x + 2y = 5, \\ 3x - y = 4. \end{cases}$$

2. Starting with rows $[1 \ 2 \mid 5]$ and $[3 \ -1 \mid 4]$, apply $R_2 \leftarrow R_2 - 3R_1$. Write the new second row.

3. What does an augmented row $[0 \ 0 \mid 3]$ tell you about the system?

4. In a system with more variables than pivot rows, what can a zero row $[0 \ 0 \ 0 \mid 0]$ indicate about the solution set?

Module 10: Matrices and Gaussian Elimination – Weekly Check-In

Name:

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Three questions. Translate freely between systems, augmented matrices, and row operations.

1. Gaussian elimination.

Consider

$$\begin{cases} x + y - z = 2, \\ 2x - y + z = 4, \\ 3x + 2y + z = 9. \end{cases}$$

- (a) Write the augmented matrix.

- (b) Apply $R_2 \leftarrow R_2 - 2R_1$ and $R_3 \leftarrow R_3 - 3R_1$, and write the resulting matrix.

- (c) Continue using elementary row operations to solve the system. State the final ordered triple.

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2. Classifying echelon forms.

For each augmented matrix, classify the system as having a unique solution, no solution, or infinitely many solutions. Briefly explain what feature of the matrix determines the classification.

$$A = \left[\begin{array}{cc|c} 1 & 2 & 4 \\ 0 & 1 & 3 \end{array} \right], \quad B = \left[\begin{array}{cc|c} 1 & -1 & 2 \\ 0 & 0 & 5 \end{array} \right],$$

$$C = \left[\begin{array}{ccc|c} 1 & 2 & -1 & 4 \\ 0 & 1 & 3 & 2 \\ 0 & 0 & 0 & 0 \end{array} \right].$$

3. Quadratic fitting by a matrix.

A quadratic $y = ax^2 + bx + c$ passes through $(0, 1)$, $(1, 4)$, and $(2, 9)$.

- Write the three linear equations for a, b, c and the corresponding augmented matrix.
- Solve for a, b, c and write the quadratic function.